Reliability: Technical Worksheet

Grade 9 · Worksheet · About 25 minutes
Work with pen and paper. Use color pencils if you like. You do not need a computer or the internet.

Name: _____

Grade/Section: _____

Date: _____

> **How to use this worksheet**
> - Do the parts **in order** from top to bottom.
> - Each part has a **time** so you know how long to spend.
> - Read every instruction before you write. You can do this on your own.

Part 1 – What you will learn (about 3 minutes)

What you will learn: Reliability means a system does what users need, when they need it, for as long as expected. You will explore **uptime**, **redundancy** (backup parts or paths), and common **failure modes**—ways systems can stop working—so you can reason about dependable digital systems.

Do this:

1. **Name one system** you rely on every school day (Wi-Fi, bus schedule app, classroom projector, shared drive).

   _____

2. In one short phrase, what goes wrong when that system is **unreliable**?

   _____

Part 2 – Uptime and availability (about 5 minutes)

Uptime is the time a system is working and reachable. **Downtime** is when it is not. People often talk about availability as a percentage (e.g. “up 99% of the month”), but what matters most is whether the system is available during the hours you actually need it.

> **Real-life scenario:** The school portal was online all weekend but offline Monday 8:00–9:30 a.m. while students submit homework. Weekend uptime looked fine in a report—yet the outage hit the most important window.

Discuss with a partner or reflect in writing: Why can high average uptime still feel unreliable to users?

Do this: Label each statement **U** (supports uptime/availability thinking) or **N** (not about uptime).

Statement	U or N?
The printer was reachable for 22 of 24 school hours yesterday	_____
The app’s icons look more modern after a redesign	_____
Login failed for the first period three mornings this week	_____
Teachers prefer a blue theme over green	_____

Part 3 – Redundancy: backup paths and spare parts (about 5 minutes)

Redundancy means having an extra path or spare so one failure does not stop everything. Examples: a second Wi-Fi access point, a spare charger, dual internet links, or saving a file both on a laptop and in a cloud folder. Redundancy costs more (money, space, setup) but raises reliability.

> **Real-life scenario:** Lab A has one network switch. When it fails, every computer loses the internet. Lab B has two switches; half the room stays online if one fails.

Critical thinking: Which lab is designed for higher reliability, and what trade-off does Lab B accept?

Do this: Match each idea to **Redundant** or **Single point of failure** (one part whose loss stops the whole service).

Setup	Type
One shared USB drive holds every club's only copy of videos	_____
Two projectors so class continues if one bulb dies	_____
Homework saved only on one phone with no backup	_____
School Wi-Fi plus a known mobile-hotspot plan for emergencies	_____

Part 4 – Failure modes (about 5 minutes)

A **failure mode** is a specific way something can break or stop working: power loss, overheating, cable unplugged, software crash, full storage, forgotten password, or slow overload. Listing likely failure modes helps you plan prevention (care, updates, backups) instead of being surprised.

> **Real-life scenario:** A classroom Chromebook cart worked for months. One morning half the devices are dead: the cart's charging strip was unplugged overnight (power failure mode). Another day a device won't open files because storage is full (capacity failure mode).

Do this:

1. **List three failure modes** for a shared classroom laptop (be specific).
   a. _____
   b. _____
   c. _____
2. **For one failure mode**, write one simple prevention step.

   _____

Part 5 – Reliability sketch (about 5 minutes)

Do this:

1. **Sketch a simple reliable setup** for one school service (Wi-Fi, file storage, or printing). Show the main path and at least one redundant/backup idea. Label a failure mode you are guarding against.

2. **Explain** in one or two sentences how your sketch improves uptime or recovery compared with a single-path design.

   _____

   _____

Part 6 – Check what you learned (about 2 minutes)

Checklist. Put a ✓ in the box when it is true for you.

☐ I can explain **reliability** in everyday language.

☐ I can describe **uptime** and why timing of downtime matters.

☐ I can give an example of **redundancy** vs a single point of failure.

☐ I can list common **failure modes** and a prevention step.

☐ I can sketch a more reliable setup with a backup path.

When all boxes are checked, you have reviewed the main ideas from this worksheet.

Worksheet on Reliability (Technical). Central topic: Reliability. Main technology area: Dependable systems and uptime.